Lattice Theory and Hard Problems

– Lattices, Geometry of Numbers, and Cryptography –

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PQC Hard Problem Colloquium

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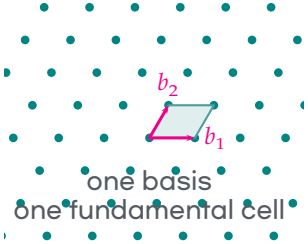
I. Introduction

I. Introduction
Discrete Geometry



A lattice is a collection of periodic points.

- It is discrete, but it fills all directions of a vector space.
- It has algebraic data: a basis and integer coefficients.
- It has geometric data: length, angle, volume, and distance.



Question. How hard is it to find the closest or shortest lattice vector?

I. Introduction

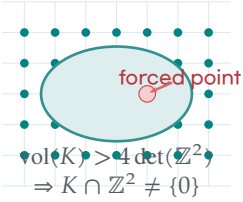
Why Lattices Matter: Geometry of Numbers

Q. What concrete problem does it solve?

Example. In $\mathbb{Z}^2$, Minkowski's theorem says that any centrally symmetric convex region with area > 4 contains a nonzero integer point. The statement is geometric, but the conclusion is arithmetic: an integer solution must exist.

What it reveals. A lattice point is not found by solving equations one-by-one; it is forced by volume. For a general lattice, the threshold is $2^n \det(\mathcal{L})$.

Why important. This is the basic move of lattice theory: prove the existence of number-theoretic objects by showing that Euclidean space has no room to avoid them.



I. Introduction

Why Lattices Matter: Quadratic Forms

Q. What is the example?

Example. The hexagonal lattice has basis

$$b_1 = (1, 0), \qquad b_2 = (1/2, \sqrt{3}/2).$$

For $z = (x, y) \in \mathbb{Z}^2$,

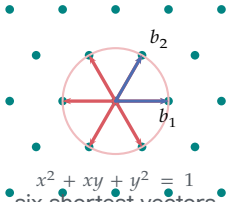
$$\|xb_1 + yb_2\|^2 = x^2 + xy + y^2.$$

What it reveals. The shortest nonzero vectors are exactly the integer solutions of

$$x^2 + xy + y^2 = 1.$$

There are six of them, which is why each disk in the hexagonal packing touches six neighbors.

Why important. A quadratic form is not just algebraic notation; it is the length function of a lattice written in integer coordinates.



I. Introduction

Why Lattices Matter: Diophantine Approximation



Q. How does approximation become a lattice problem?

Example. To approximate π, look for integers p, q with $|p - q\pi|$ tiny. The famous approximation

$$\frac{355}{113} \approx \pi$$

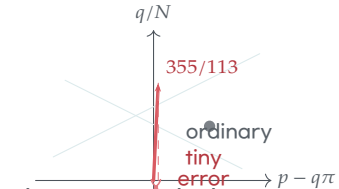
means that the integer vector $(p, q) = (355, 113)$ makes the linear form $p - q\pi$ unusually small.

Lattice viewpoint. For a scale N, vectors of the form

$$(p - q\pi,\ q/N)$$

balance two demands: small error and not-too-large denominator.

Why important. Lattices explain why good approximations are short vectors: the numerator, denominator, and error become one geometric object.



I. Introduction

Why Lattices Matter: Sphere Packing



Q. What is the concrete packing example?

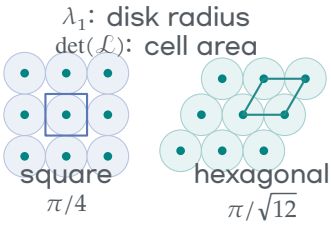
Example. In the square lattice $\mathbb{Z}^2$, unit spacing gives disk density

$$\Delta(\mathbb{Z}^2) = \frac{\pi(1/2)^2}{1} = \frac{\pi}{4} \approx .785.$$

In the hexagonal lattice, the same nearest-neighbor distance gives

$$\Delta = \frac{\pi}{\sqrt{12}} \approx .907.$$

What it reveals. The key quantities are the shortest vector length $\lambda_1(\mathcal{L})$ and the cell area $\det(\mathcal{L})$.
Why important. The best two-dimensional packing is not discovered by drawing many circles; it is explained by the geometry of one lattice cell and its shortest vectors.



I. Introduction

Why Lattices Matter: Integer Programming

Q. Where does lattice geometry enter optimization?

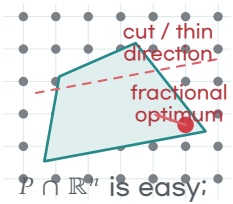
Example. An integer program asks for a point in

$$P \cap \mathbb{Z}^n,$$

where P is a polytope. Its linear relaxation searches $P \cap \mathbb{R}^n$ instead. A fractional optimum may sit near the middle of a face, while every nearby integer point violates a constraint.

What it reveals. Rounding is a geometric question: how is the real feasible region positioned relative to the lattice $\mathbb{Z}^n$? If P is thin in some integer direction, that direction is the useful branch or cut.

Why important. Lattice width, flatness, and rounding explain why continuous optimization is only the beginning of integer optimization.



$P \cap \mathbb{R}^n$ is easy;
$P \cap \mathbb{Z}^n$ depends on the grid

I. Introduction
Why Lattices Matter: LLL Reduction



Q. What does LLL actually uncover?

Example. The vectors

$$c_1 = (1001, 1000), \qquad c_2 = (1000, 999)$$

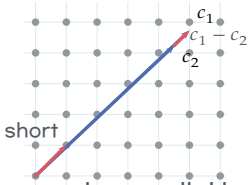
form a basis of $\mathbb{Z}^2$ because $\det[c_1\ c_2] = -1$. Both basis vectors have length about 1414, but

$$c_1 - c_2 = (1, 1)$$

is very short.

What it reveals. A basis can hide the geometry of its own lattice. LLL performs integer column operations to expose shorter, more orthogonal directions.

Why important. Many attacks and algorithms begin by asking: is the secret simply hidden by a bad basis? LLL is the first systematic test.



long nearly-parallel basis
hides a short difference

I. Introduction

Why Lattices Matter: Post-Quantum Cryptography

Q. What is the concrete cryptographic shape?

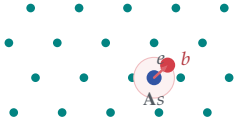
Example. In LWE, the public data has the form

$$b = \mathbf{A}s + e \pmod{q},$$

where s is secret and e is small noise. Without e, this is ordinary linear algebra modulo q. With e, it becomes a noisy decoding problem.

Lattice viewpoint. After lifting modulo q to integers, b is close to a structured q-ary lattice point, but the small error hides which one.

Why important. Known quantum algorithms are excellent at exact algebraic structure; lattice cryptography deliberately adds controlled noise and shortness constraints.



without noise:
linear algebra
with noise: decoding

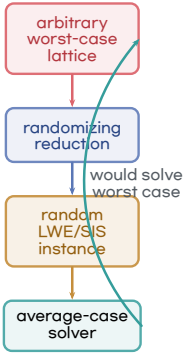
I. Introduction

Why Lattices Matter: Worst-case to Average-case



Q. Why is this stronger than a random puzzle assumption?

Example. A cryptographic public key should look random, so an attacker sees an average-case LWE or SIS instance, not a handcrafted worst-case lattice.
Reduction idea. For suitable parameters, an algorithm that solves random LWE/SIS instances would imply algorithms for approximate GapSVP/SIVP on arbitrary lattices. The random instance is tied back to the worst lattice geometry.
Why important. This is rare in cryptography: the security evidence for random public instances is connected to broad worst-case mathematical hardness, not only to failed attack attempts.



Note. The same object can be easy to describe by a basis and difficult to understand geometrically.

I. Introduction
The Dictionary



Algebra	Geometry
Basis matrix $\mathbf{B}$	Skew coordinate grid in $\mathbb{R}^m$
Integer vector $\mathbf{z} \in \mathbb{Z}^n$	Lattice point $\mathbf{Bz}$
Unimodular change $\mathbf{U} \in \mathrm{GL}_n(\mathbb{Z})$	Same lattice, different basis
$\lvert \det \mathbf{B} \rvert$	Volume of a fundamental cell
Dual basis $\mathbf{B}^{-T}$	Linear tests taking integer values

Warning. A good coordinate system is not automatic. A bad basis can hide very short vectors.

I. Introduction
Roadmap



1. Define lattices, bases, determinant, and the dual lattice.
2. Use geometry of numbers to prove that short vectors must exist.
3. State the main computational lattice problems.
4. Explain why SIS and LWE are lattice problems in disguise.



II. Lattices

II. Lattices
Definition



Lattice generated by a basis

Let $b_1, \dots, b_n \in \mathbb{R}^m$ be linearly independent. The lattice generated by them is

$$\mathcal{L}(\mathbf{B}) = \left\{ \sum_{i=1}^{n} z_i b_i : z_i \in \mathbb{Z} \right\} = \mathbf{B}\mathbb{Z}^n,$$

where $\mathbf{B} = [b_1 \ \cdots \ b_n] \in \mathbb{R}^{m \times n}$.

- The rank is n.
- The ambient dimension is m.
- A full-rank lattice in $\mathbb{R}^n$ has $m = n$.

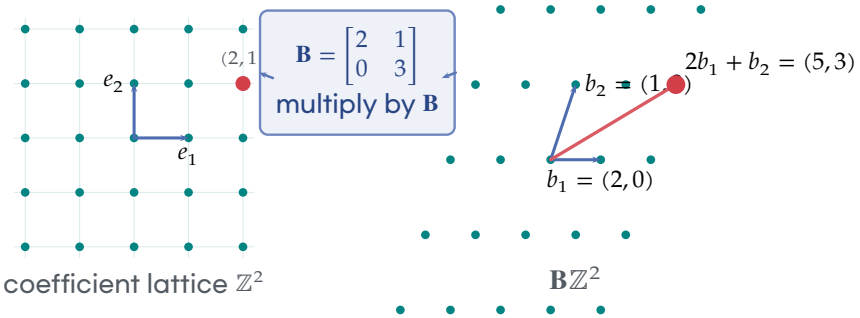
Equivalent viewpoint. A lattice is a discrete additive subgroup of $\mathbb{R}^m$.
Concrete example. With $b_1 = (2,0)$ and $b_2 = (1,3)$,

$$\mathcal{L} = \{u(2,0) + v(1,3) : u, v \in \mathbb{Z}\} = \{(2u + v, 3v) : u, v \in \mathbb{Z}\}.$$

Thus $(4,6) = 1b_1 + 2b_2 \in \mathcal{L}$, while $(1,0) \notin \mathcal{L}$.

II. Lattices

Visualization: Integer Coordinates Become Geometry



$$(u, v) \in \mathbb{Z}^2 \quad \longmapsto \quad ub_1 + vb_2 = (2u + v, 3v).$$

Point. A lattice basis is a linear machine that turns integer coordinates into a discrete geometric point set.

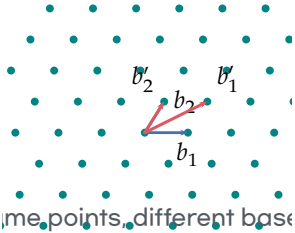
II. Lattices
Many Bases, One Lattice

Change of basis

If $\mathbf{B}$ is a basis matrix and $\mathbf{U} \in \mathrm{GL}_n(\mathbb{Z})$ is unimodular, then

$$\mathcal{L}(\mathbf{BU}) = \mathcal{L}(\mathbf{B}).$$

Conversely, two full-rank bases of the same lattice differ by such a matrix.



- Unimodular means $\mathbf{U}$ has integer entries and determinant ± 1.
- The lattice is the invariant object; the basis is only a coordinate choice.

Example. For $\mathbf{U} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$,

$$[b_1\ b_2]\mathbf{U} = [b_1\ \ b_1 + b_2].$$

The basis changes, but the point set does

II. Lattices
Fundamental Parallelepiped



Fundamental parallelepiped

For a basis $\mathbf{B} = [b_1 \ \cdots \ b_n]$ of a full-rank lattice,

$$\mathcal{P}(\mathbf{B}) = \left\{ \sum_{i=1}^{n} t_i b_i : 0 \le t_i < 1 \right\}.$$

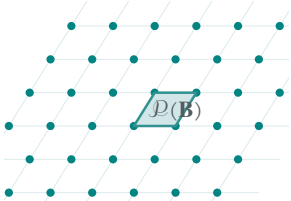
Every $x \in \mathbb{R}^n$ has a unique decomposition $x = v + p$ with $v \in \mathcal{L}$ and $p \in \mathcal{P}(\mathbf{B})$.

- $\mathcal{P}(\mathbf{B})$ tiles the space by lattice translations.
- It is a coordinate-dependent fundamental domain.

Example in $\mathbb{Z}^2$. For $\mathbf{B} = \mathbf{I}_2$,

$$(2.7, -1.3) = (2, -2) + (0.7, 0.7).$$

The first term is in $\mathbb{Z}^2$; the second lies in $[0,1)^2$.



II. Lattices
Determinant and Covolume



Determinant

For a full-rank lattice $\mathcal{L} = \mathcal{L}(\mathbf{B}) \subset \mathbb{R}^n$,

$$\det(\mathcal{L}) = |\det \mathbf{B}|.$$

Equivalently, $\det(\mathcal{L})$ is the volume of any fundamental parallelepiped.

- This number is independent of the basis:

$$|\det(\mathbf{BU})| = |\det \mathbf{B}|\,|\det \mathbf{U}| = |\det \mathbf{B}|.$$

- Small determinant means many lattice points per unit volume; large determinant means sparse lattice points.

Counting heuristic

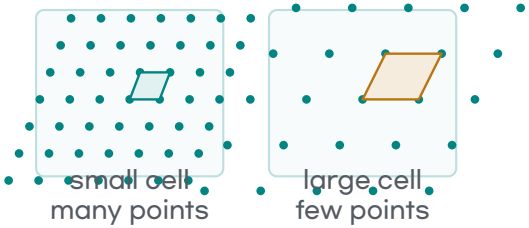
For a reasonably round region $S \subset \mathbb{R}^n$,

$$|\mathcal{L} \cap S| \approx \frac{\mathrm{vol}(S)}{\det(\mathcal{L})}.$$

Example. If $b_1 = (2,0)$ and $b_2 = (1,3)$, then $\det(\mathcal{L}) = 6$. A region of area 600 should contain roughly $600/6 = 100$ lattice points.

II. Lattices

Visualization: Covolume Measures Density



Same window, different covolume.

$$\#(\mathcal{L} \cap S) \approx \frac{\mathrm{vol}(S)}{\det(\mathcal{L})}.$$

Concrete reading. If the fundamental cell has area 6, then each lattice point occupies about six square units of plane.

II. Lattices
The Dual Lattice



Dual lattice

The dual lattice is

$$\mathcal{L}^* = \{y \in \mathrm{span}(\mathcal{L}) : \langle x, y\rangle \in \mathbb{Z} \text{ for all } x \in \mathcal{L}\}.$$

If $\mathcal{L} = \mathcal{L}(\mathbf{B})$ is full-rank, then

$$\mathcal{L}^* = \mathcal{L}(\mathbf{B}^{-T}).$$

Facts

- $(\mathcal{L}^*)^* = \mathcal{L}$.
- $\det(\mathcal{L}^*) = 1/\det(\mathcal{L})$.
- Dual vectors define integer-valued tests on $\mathcal{L}$.

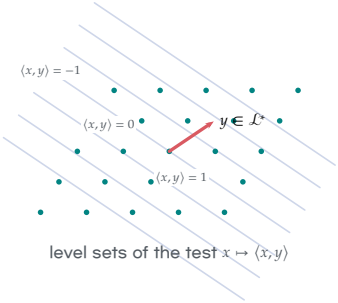
Cryptographic role

- Modular equations define q-ary lattices.
- Fourier analysis naturally sees the dual lattice.
- LWE noise is measured relative to dual constraints.

Concrete test. If $\mathcal{L} = 2\mathbb{Z} \times 3\mathbb{Z}$, then $\mathcal{L}^* = \frac{1}{2}\mathbb{Z} \times \frac{1}{3}\mathbb{Z}$. The vector $y = (1/2, 1/3)$ is dual because $\langle (2a, 3b), y\rangle = a + b \in \mathbb{Z}$.

II. Lattices

Visualization: Dual Vectors Are Integer Tests



level sets of the test $x \mapsto \langle x, y \rangle$

Interpretation. A dual vector does not merely point in space. It defines a ruler:

$$x \in \mathcal{L} \quad \Longrightarrow \quad \langle x, y \rangle \in \mathbb{Z}.$$

Concrete example. For $\mathcal{L} = 2\mathbb{Z} \times 3\mathbb{Z}$, the dual vector $y = (1/2, 1/3)$ sends $(2a, 3b)$ to $a + b$.

II. Lattices
Gram Matrix



Gram matrix

Given a basis $\mathbf{B} = [b_1 \ \cdots \ b_n]$, the Gram matrix is

$$\mathbf{G} = \mathbf{B}^T\mathbf{B}, \qquad G_{ij} = \langle b_i, b_j \rangle .$$

- $\mathbf{G}$ stores all lengths and angles of the basis vectors.
- $\det(\mathbf{G}) = (\det \mathbf{B})^2$ for full-rank square $\mathbf{B}$.
- Quadratic forms appear because

$$\|\mathbf{B}z\|^2 = z^T\mathbf{G}z, \qquad z \in \mathbb{Z}^n.$$

Key reduction

Finding a short nonzero lattice vector is equivalent to minimizing a positive definite quadratic form over $\mathbb{Z}^n \setminus \{0\}$.

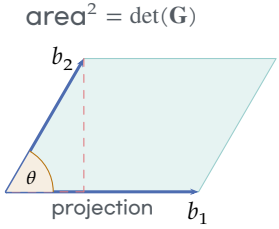
Example. For $b_1 = (2,0)$ and $b_2 = (1,3)$,

$$\mathbf{G} = \begin{bmatrix} 4 & 2 \\ 2 & 10 \end{bmatrix}, \qquad \|xb_1 + yb_2\|^2 = 4x^2 + 4xy + 10y^2.$$

II. Lattices

Visualization: The Gram Matrix Stores Angles





$$\mathbf{G} = \begin{bmatrix} \|b_1\|^2 & \langle b_1, b_2 \rangle \\ \langle b_2, b_1 \rangle & \|b_2\|^2 \end{bmatrix}.$$

Why it matters. Minimizing $\|\mathbf{B}z\|$ is the same as minimizing

$$z^T \mathbf{G} z.$$

So geometry becomes arithmetic in a positive definite quadratic form.

III. Geometry of Numbers

III. Geometry of Numbers

Shortest Vector



First minimum

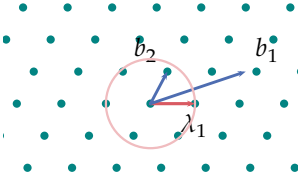
The first successive minimum of $\mathcal{L}$ is

$$\lambda_1(\mathcal{L}) = \min\{\|v\| : v \in \mathcal{L} \setminus \{0\}\}.$$

Any nonzero $v \in \mathcal{L}$ with $\|v\| = \lambda_1(\mathcal{L})$ is a shortest vector.

- λ_1 is geometric: it does not depend on the basis.
- It is hard to read from a badly skewed basis.
- Cryptography uses the gap between basis description and geometric search.

Example. For $\mathcal{L} = 2\mathbb{Z} \times 5\mathbb{Z}$, the shortest nonzero vectors are $(\pm 2, 0)$, so $\lambda_1(\mathcal{L}) = 2$.



st vector versus basis v

III. Geometry of Numbers

Successive Minima



Successive minima

For $1 \le i \le n$, the i-th successive minimum is

$$\lambda_i(\mathcal{L}) = \inf\left\{r > 0 : \dim \operatorname{span}(\mathcal{L} \cap r\mathcal{B}_2^n) \ge i\right\}.$$

- λ_i is the radius needed to find i linearly independent lattice vectors.
- The sequence is monotone:

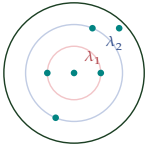
$$\lambda_1(\mathcal{L}) \le \lambda_2(\mathcal{L}) \le \cdots \le \lambda_n(\mathcal{L}).$$

- The last minimum λ_n measures how large a ball must be to reveal a full-rank lattice basis.

Example. For $\mathcal{L} = 2\mathbb{Z} \times 5\mathbb{Z} \subset \mathbb{R}^2$,

$$\lambda_1(\mathcal{L}) = 2, \qquad \lambda_2(\mathcal{L}) = 5.$$

Radius 2 finds one direction; radius 5 finds two independent directions.



III. Geometry of Numbers

Minkowski's Convex Body Theorem



Minkowski

Let $\mathcal{L} \subset \mathbb{R}^n$ be a full-rank lattice and let $K \subset \mathbb{R}^n$ be convex, centrally symmetric, and measurable. If

$$\mathrm{vol}(K) > 2^n \det(\mathcal{L}),$$

then K contains a nonzero lattice point.

Meaning

- Large symmetric bodies cannot avoid all nonzero lattice points.
- Volume alone forces arithmetic structure.

Why the factor 2^n?

- Compare $K/2$ with a fundamental domain.
- If two points of $K/2$ are congruent modulo $\mathcal{L}$, their difference lies in $K \cap \mathcal{L}$.

Example in $\mathbb{Z}^2$. Any centrally symmetric convex region with area > 4 contains a nonzero integer point. A long thin diamond can force an integer relation even when no coordinate is individually small.

III. Geometry of Numbers

A Short Vector Always Exists



Minkowski bound for λ_1

For every full-rank lattice $\mathcal{L} \subset \mathbb{R}^n$,

$$\lambda_1(\mathcal{L}) \leq \sqrt{n}\, \det(\mathcal{L})^{1/n}.$$

Proof idea.

- Apply Minkowski's theorem to a cube centered at the origin.
- Choose side length slightly larger than $2\det(\mathcal{L})^{1/n}$.
- The cube contains a nonzero lattice point v with

$$\|v\|_2 \leq \sqrt{n}\, \det(\mathcal{L})^{1/n}.$$

Important. Existence is not an efficient algorithm for finding such a vector.
Example. In dimension 2, if $\det(\mathcal{L}) = 25$, Minkowski gives

$$\lambda_1(\mathcal{L}) \leq \sqrt{2} \cdot 5 \approx 7.07.$$

So a nonzero vector of length at most 7.07 must exist, even if the displayed basis is much longer.

III. Geometry of Numbers

Gaussian Heuristic



Heuristic estimate

The expected number of lattice points in a ball of radius r is approximately

$$\frac{\mathrm{vol}(r\mathcal{B}_2^n)}{\det(\mathcal{L})}.$$

The Gaussian heuristic predicts $\lambda_1(\mathcal{L})$ by setting this ratio near 1.

$$\mathrm{vol}(r\mathcal{B}_2^n) = \frac{\pi^{n/2}}{\Gamma(n/2+1)} r^n.$$

- Random-looking lattices often behave close to this estimate.
- Cryptographic parameter choices compare attacks against this geometry.
- The heuristic is not a theorem for every lattice.

Example. In $\mathbb{R}^2$ with $\det(\mathcal{L}) = 100$, the heuristic expects one lattice point in a disk when

$$\pi r^2/100 \approx 1, \qquad r \approx \sqrt{100/\pi} \approx 5.64.$$

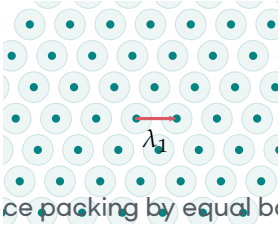
This is a prediction, not a guarantee.

III. Geometry of Numbers

Sphere Packing Viewpoint



Place a ball of radius $\lambda_1(\mathcal{L})/2$ around each lattice point. These balls do not overlap.



$$\Delta(\mathcal{L}) = \frac{\mathrm{vol}((\lambda_1(\mathcal{L})/2)\mathcal{B}_2^n)}{\det(\mathcal{L})}$$

is the density of the corresponding lattice sphere packing. **Example.** For $\mathbb{Z}^2$, $\lambda_1 = 1$ and $\det = 1$, so the packing density is $\pi(1/2)^2 = \pi/4$. Changing the lattice shape to the hexagonal lattice improves the density.

III. Geometry of Numbers

Hermite Constant



Hermite invariant

For a full-rank lattice $\mathcal{L} \subset \mathbb{R}^n$,

$$\gamma(\mathcal{L}) = \frac{\lambda_1(\mathcal{L})^2}{\det(\mathcal{L})^{2/n}}.$$

The Hermite constant is

$$\gamma_n = \sup_{\mathcal{L} \subset \mathbb{R}^n} \gamma(\mathcal{L}).$$

- It asks for the densest possible short-vector behavior in dimension n.
- It is closely tied to optimal lattice sphere packings.
- Computational lattice reduction is, in part, an attempt to find bases witnessing small vectors.

Dimension 2. The square lattice has $\gamma(\mathbb{Z}^2) = 1$. The hexagonal lattice has $\lambda_1 = 1$ and $\det = \sqrt{3}/2$, hence

$$\gamma = \frac{1}{\sqrt{3}/2} = \frac{2}{\sqrt{3}} = \gamma_2.$$

It is optimal in the plane.

IV. Hard Problems

IV. Hard Problems
SVP



Shortest Vector Problem

Given a basis $\mathbf{B}$ of a lattice $\mathcal{L}$, find a nonzero vector $v \in \mathcal{L}$ such that

$$\|v\| = \lambda_1(\mathcal{L}).$$

Approximate SVP

$$\|v\| \le \gamma(n)\lambda_1(\mathcal{L}).$$

- Exact SVP is too strong for many applications.
- Approximation factors encode the security target.

Basis trap

- A basis vector may be exponentially longer than λ_1.
- Short combinations can have large integer coefficients in a bad basis.

Toy instance. For $\mathcal{L} = 2\mathbb{Z} \times 5\mathbb{Z}$, SVP returns $(\pm 2, 0)$ and $\lambda_1 = 2$. With $c_1 = (1001, 1000)$, $c_2 = (1000, 999)$, the short vector is hidden as $c_1 - c_2 = (1, 1)$.

IV. Hard Problems
CVP



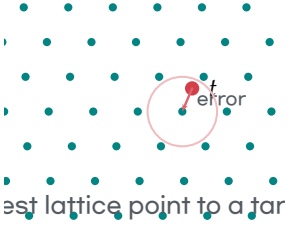
Closest Vector Problem

Given a basis of $\mathcal{L} \subset \mathbb{R}^n$ and a target $t \in \mathbb{R}^n$, find $v \in \mathcal{L}$ minimizing

$$\|t - v\|.$$

- CVP asks for the lattice point in the same Voronoi cell as t.
- Bounded Distance Decoding (BDD) is CVP with a promise that t is close to the lattice.
- Decoding linear codes and decoding lattices are parallel ideas.

Example in $\mathbb{Z}^2$. For $t = (2.2, 4.7)$, the closest point is $(2, 5)$. For a skew lattice, nearest-point search is no longer coordinate-wise rounding.



IV. Hard Problems
SIVP and GapSVP



SIVP

Given a basis of $\mathcal{L}$, find n linearly independent vectors

$$v_1, \dots, v_n \in \mathcal{L}$$

with $\max_i \|v_i\| \le \gamma(n)\lambda_n(\mathcal{L})$.

GapSVP

Given a threshold d, distinguish between

$$\lambda_1(\mathcal{L}) \le d \quad \text{and} \quad \lambda_1(\mathcal{L}) > \gamma(n)d.$$

Why they appear in reductions

- SIVP captures the need for many independent short directions.
- GapSVP turns search geometry into a decision problem.
- Worst-case reductions often target approximate versions of these problems.

Example. For $\mathcal{L} = 2\mathbb{Z} \times 5\mathbb{Z}$, SIVP needs $(2,0)$ and $(0,5)$. GapSVP with $d = 3$ and $\gamma = 2$ asks whether $\lambda_1 \le 3$ or $\lambda_1 > 6$.

IV. Hard Problems
Naive Algorithms



Enumeration

- Search all integer vectors z with $\|\mathbf{B}z\| \leq R$.
- Works in low dimensions.
- The search tree grows exponentially with the dimension.

Voronoi methods

- Describe the Voronoi cell and use its facets.
- Exact but exponential.

Lesson. Exact lattice problems are feasible for small n and become rapidly expensive. **Enumeration scale.** If $z_i \in [-10, 10]$, brute force checks 21^n candidates: 441 in dimension 2, but about 1.7×10^{13} in dimension 10.

IV. Hard Problems
Visualization: Enumeration Explosion



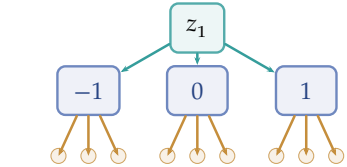
Coefficient search. Enumeration searches integer coordinates first:

$$z = (z_1, \dots, z_n), \qquad \|\mathbf{B}z\| \le R.$$

If each coordinate has $2R+1$ choices, the raw search width is

$$(2R+1)^n.$$

The geometry filters candidates, but the tree is still exponential.



IV. Hard Problems
LLL Reduction



Goal of basis reduction

Replace a bad basis by a better basis of the same lattice:

$$\mathbf{B} \longmapsto \mathbf{BU}, \qquad \mathbf{U} \in \mathrm{GL}_n(\mathbb{Z}).$$

LLL theorem, informal

The LLL algorithm runs in polynomial time and outputs a basis whose first vector satisfies

$$\|b_1\| \le 2^{(n-1)/2}\lambda_1(\mathcal{L}).$$

- LLL is powerful in practice but gives an exponential approximation factor.
- Stronger block reduction algorithms trade more time for better quality.
- Lattice cryptography chooses parameters beyond known reduction attacks.

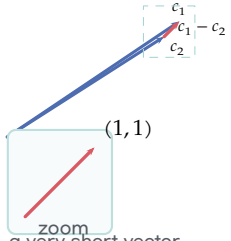
Concrete effect. Starting from $c_1 = (1001, 1000)$ and $c_2 = (1000, 999)$, integer subtraction exposes

$$c_1 - c_2 = (1, 1).$$

LLL automates this kind of basis improvement in high dimension.

IV. Hard Problems

Visualization: LLL Finds Integer Cancellations



a very short vector
can be a difference
of long basis vectors

$$c_1 = (1001, 1000), \qquad c_2 = (1000, 999).$$

$$c_1 - c_2 = (1, 1).$$

What reduction does. It repeatedly replaces a vector by an integer combination that exposes shorter directions in the same lattice.

IV. Hard Problems

Penetrating Example: Subset Sum I



Subset sum

Given weights $a_1, \dots, a_n$ and target T, find $x \in \{0,1\}^n$ such that

$$\sum_{i=1}^{n} a_i x_i = T.$$

Lattice embedding. For a large scale M, generate a lattice in $\mathbb{R}^{n+1}$ by

$$(e_i,\ Ma_i) \quad (1 \le i \le n), \qquad (0, \dots, 0,\ -MT).$$

If x is a solution, the lattice contains

$$(x_1, \dots, x_n,\ M(\sum a_i x_i - T)) = (x_1, \dots, x_n, 0),$$

which is unusually short.

Why this is penetrating. The combinatorial condition "exactly hit T" is converted into the geometric condition "make the last coordinate zero." Large M makes wrong sums visibly long.

IV. Hard Problems

Penetrating Example: Subset Sum II



Concrete instance.

$$\begin{aligned} a &= (366, 385, 392, 401, \\ &\quad 422, 437, 450, 463), \\ T &= 1204. \end{aligned}$$

$$366 + 401 + 437 = 1204.$$

So LLL is asked to find the short vector

$$(1, 0, 0, 1, 0, 1, 0, 0, 0).$$

What LLL sees. The vector has length $\sqrt{3}$, while a wrong subset with sum error $r \neq 0$ has last coordinate Mr.

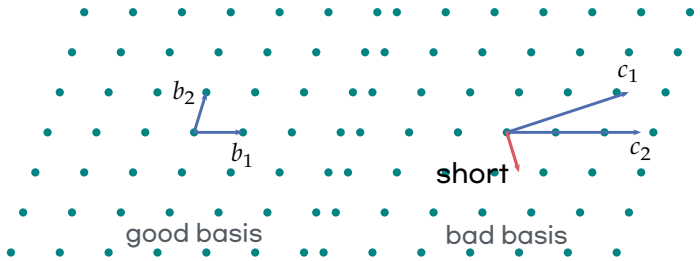
For $M = 100$, even a one-unit target miss contributes length at least 100.



Point. A discrete search problem becomes a geometric question: is the hidden $0/1$ vector among the shortest lattice vectors?

IV. Hard Problems

Good Bases and Bad Bases



A lattice basis is like a coordinate system. The point set is unchanged, but the computational difficulty can be completely different.

$$(1001, 1000) - (1000, 999) = (1, 1)$$

Concrete lesson. A short vector may appear only as a cancellation between long basis vectors.

IV. Hard Problems
Complexity Landscape



Problem	Exact version	Approximate version
SVP	NP-hard under randomized reductions for some norms	Central for security estimates
CVP	At least as hard as SVP in many settings	BDD supports decoding-style cryptography
SIVP	Worst-case geometric problem	Used in worst-case to average-case reductions
GapSVP	Decision form of short-vector geometry	Often paired with SIVP

Caution. Cryptographic assumptions use concrete approximation regimes, not just the slogan "SVP is hard." **Parameter meaning.** Exact SVP means $\gamma = 1$. Cryptographic estimates ask what block size is needed for a concrete approximation factor.

V. Cryptographic Lattices

V. Cryptographic Lattices

q-ary Lattices



q-ary lattice from a matrix

Let $\mathbf{A} \in \mathbb{Z}_q^{m \times n}$. Define

$$\Lambda_q^{\perp}(\mathbf{A}) = \{x \in \mathbb{Z}^n : \mathbf{A}x \equiv 0 \pmod{q}\}.$$

This is an integer lattice containing $q\mathbb{Z}^n$.

- Modular linear algebra becomes lattice geometry after lifting to $\mathbb{Z}^n$.
- The determinant is typically a power of q when $\mathbf{A}$ has full rank modulo q.
- Short integer relations modulo q are short lattice vectors.

Toy example. Take $q = 5$ and $\mathbf{A} = [1 \;\; 2]$. Then

$$\Lambda_5^{\perp}(\mathbf{A}) = \{(x_1, x_2) \in \mathbb{Z}^2 : x_1 + 2x_2 \equiv 0 \pmod{5}\}.$$

The vector $(3, 1)$ lies in this lattice because $3 + 2 = 5$.



V. Cryptographic Lattices

SIS



Short Integer Solution

Given a random matrix $\mathbf{A} \in \mathbb{Z}_q^{m \times n}$, find a nonzero short vector $x \in \mathbb{Z}^n$ such that

$$\mathbf{A}x \equiv 0 \pmod{q}.$$

Lattice form

$$x \in \Lambda_q^{\perp}(\mathbf{A}), \qquad 0 < \|x\| \le \beta.$$

SIS is a short-vector problem in a q-ary lattice. **Toy relation.** For $\mathbf{A} = [1 \;\; 2] \bmod 5$, $x = (3, 1)$ satisfies $\mathbf{A}x = 5 \equiv 0$. Real SIS uses dimensions where finding such short relations is hard.

Cryptographic intuition

- Easy to verify: compute $\mathbf{A}x \bmod q$.
- Hard to find when parameters make short relations rare.
- Gives collision-resistant hashing and signatures.

V. Cryptographic Lattices

Visualization: SIS as Congruence Lines



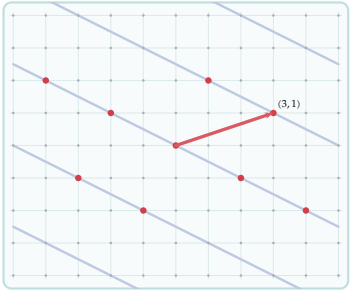
Toy kernel. For $q = 5$ and $\mathbf{A} = [1 \;\; 2]$,

$$x_1 + 2x_2 \equiv 0 \pmod 5.$$

The lattice is the union of integer points on the parallel lines

$$x_1 + 2x_2 = 5k, \qquad k \in \mathbb{Z}.$$

SIS asks for a short nonzero point on these lines.



red points satisfy
$x_1 + 2x_2 \equiv 0 \bmod 5$

V. Cryptographic Lattices

LWE



Learning With Errors

Let $s \in \mathbb{Z}_q^n$ be secret. Given samples

$$(a_i,\ b_i = \langle a_i, s\rangle + e_i \bmod q),$$

where $a_i \leftarrow \mathbb{Z}_q^n$ and e_i is small, recover s or distinguish the samples from uniform.

- Without error, this is linear algebra over $\mathbb{Z}_q$.
- With small error, the equations are only approximately consistent.
- The error hides the exact lattice hyperplanes that would reveal s.

Core idea. Noise turns an easy algebra problem into a closest-vector style problem. **One sample.** Let $q = 17$, $a = (3, 5)$, $s = (4, 2)$, and $e = -1$. Then

$$b = \langle a, s\rangle + e = 3 \cdot 4 + 5 \cdot 2 - 1 = 21 \equiv 4 \pmod{17}.$$

Without the error, the value would be 5; the small shift hides the exact equation.

V. Cryptographic Lattices

Visualization: LWE Thickens Exact Equations

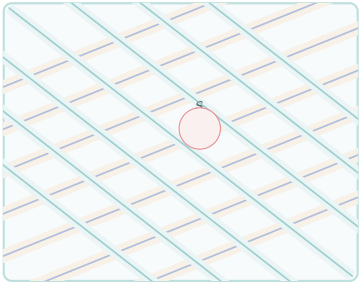


No error. Each equation cuts the secret space by a thin modular hyperplane.

With error. The hyperplane becomes a strip:

$$\langle a_i, s \rangle \approx b_i \pmod{q}.$$

Many strips intersect near the secret, but the exact intersection is blurred.



error turns exact hyperplanes into noisy strips

V. Cryptographic Lattices

LWE as Bounded Distance Decoding



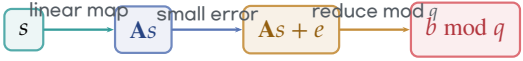
Collect m samples into a matrix $\mathbf{A} \in \mathbb{Z}_q^{m \times n}$ and vector

$$b = \mathbf{A}s + e \pmod{q}.$$

Decoding view

The vector b is close to the code/lattice generated by columns of $\mathbf{A}$ modulo q. Recovering s means identifying the nearby structured point despite the error e.

Decoding analogy. In the toy sample above, 4 is close to the noiseless value 5 modulo 17. BDD asks for the hidden structured point when many coordinates have small errors.



V. Cryptographic Lattices

Penetrating Example: Primal LWE Attack I



LWE as BDD

Given $\mathbf{A} \in \mathbb{Z}_q^{m \times n}$ and $b = \mathbf{A}s + e \pmod q$, define the q-ary lattice

$$\Lambda_q(\mathbf{A}) = \{\mathbf{A}z + qy : z \in \mathbb{Z}^n,\ y \in \mathbb{Z}^m\} \subseteq \mathbb{Z}^m.$$

Then $\mathbf{A}s \in \Lambda_q(\mathbf{A})$ and b is close to it by the error vector e.

Attack question. The attacker sees $\mathbf{A}$ and b, but not s or e:

$$\text{find the closest point of } \Lambda_q(\mathbf{A}) \text{ to } b.$$

Once that point is recovered as $\mathbf{A}s$, solving for s is linear algebra modulo q.
Why this is penetrating. LWE is not just "linear equations with noise." It is a decoding problem in a high-dimensional modular lattice where the noise radius is deliberately below the expected lattice spacing.

V. Cryptographic Lattices

Penetrating Example: Primal LWE Attack II



Concrete sample. Let $q = 17$,

$$\mathbf{A} = \begin{bmatrix} 3 & 5 \\ 7 & 4 \\ 2 & 9 \end{bmatrix}, \qquad s = \begin{bmatrix} 4 \\ 2 \end{bmatrix}, \qquad e = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}.$$

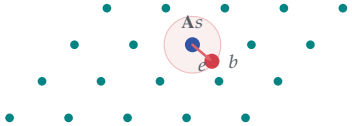
Then

$$\mathbf{A}s \equiv (5, 2, 9), \qquad b = (6, 1, 9) \pmod{17}.$$

Geometric meaning.

$$b - (\mathbf{A}s) = e, \qquad \|e\| = \sqrt{2}.$$

The public vector b is a point near the hidden lattice point $\mathbf{A}s$. The error vector is the displacement that hides the exact equation.



LWE attack surface: BDD in a q-ary lattice

V. Cryptographic Lattices

Trapdoors



Lattice trapdoor, informal

A public basis may be bad for solving CVP or sampling short vectors, while a secret basis of the same lattice is good.

Public view

- Random-looking modular matrix.
- Hard basis for the associated lattice.
- Short preimages look difficult to find.

Secret view

- Structured short basis.
- Efficient Gaussian sampling.
- Enables signatures and identity-based encryption.

Security shape

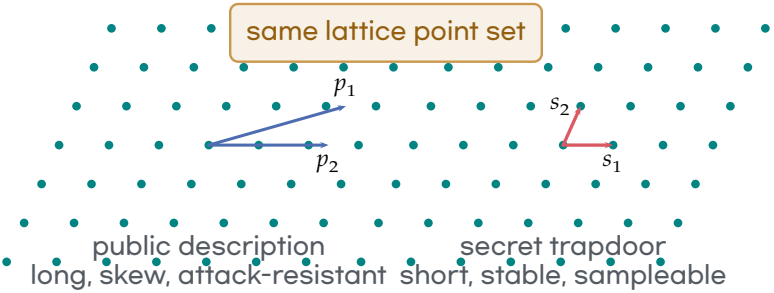
The public key describes the lattice; the trapdoor describes it well.

Example shape. The public matrix may define the same lattice as a secret short basis. To outsiders it resembles a bad basis; to the owner it resembles a basis exposing short vectors.

V. Cryptographic Lattices

Visualization: Public Basis, Secret Basis





Meaning. A trapdoor is not a different lattice; it is a better hidden coordinate system for the same public lattice.

V. Cryptographic Lattices

Worst-case to Average-case



Ajtai-Regev philosophy

Solving certain random lattice problems on average would imply algorithms for hard lattice problems in the worst case.

- SIS is connected to worst-case SIVP and related short-vector problems.
- LWE is connected to worst-case GapSVP and SIVP under suitable parameters.
- This is stronger evidence than a purely ad hoc average-case assumption.

Concrete reading. A solver for random-looking LWE samples would become a solver for geometric problems on arbitrary lattices.



V. Cryptographic Lattices

What Parameters Control



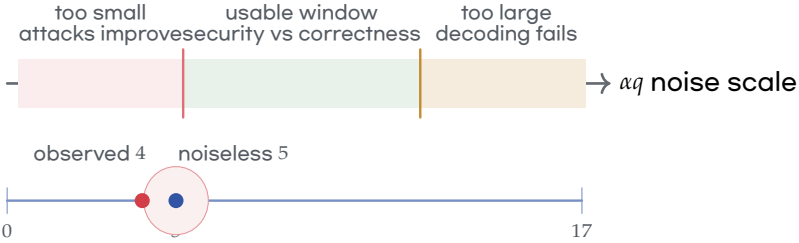
Parameter	Role
n	Main security dimension; larger n makes lattice attacks harder.
q	Modulus; controls algebraic space and noise wraparound.
m	Number of samples or equations; affects density and reductions.
αq	Error scale in LWE; must hide the secret without destroying correctness.
β	Shortness bound in SIS; controls collision resistance and extraction.

Design tension. Correctness wants small noise and short vectors; security wants them not too small. **Toy tension.** With $q = 17$, error in $\{-1, 0, 1\}$ is easy to decode if the secret structure is known. Too-large errors break correctness; too-small errors help attacks.

V. Cryptographic Lattices

Visualization: Parameter Tradeoff





toy $q = 17$: noise must hide the equation without wrapping into ambiguity

V. Cryptographic Lattices

Summary



- A lattice is a discrete additive subgroup of Euclidean space.
- Bases are algebraic descriptions; lengths and volumes are geometric invariants.
- Minkowski's theorem proves short vectors exist, but does not find them efficiently.
- SVP, CVP, SIVP, and GapSVP formalize the hard search and decision tasks.
- SIS and LWE turn modular linear algebra plus shortness/noise into lattice hardness.

One-line takeaway

Lattice cryptography is built on the gap between compact algebraic descriptions and difficult Euclidean search.

Examples to remember. Minkowski: area > 4 forces an integer point in $\mathbb{Z}^2$. LLL: $(1001, 1000) - (1000, 999) = (1, 1)$ exposes hidden shortness. LWE: $b = \mathbf{A}s + e \bmod q$ turns exact linear algebra into noisy decoding.

Thank You

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